

A-level

ENVIRONMENTAL SCIENCE

Questions by topic: Solid wastes - Set 1

Mark scheme

AQA spec 3.4.3.2.12; Pollution; textbook pp. 279-285

Marking guidance

- The number in brackets after each marking point is the mark it earns; the note above the points says how many are needed.
- Where a question asks for a fixed number of points (eg *two reasons*), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	<i>1 mark for each point.</i> • Landfill fell from 79% to 14% (large decrease) (1) • Incineration rose from 9% to 42% and recycling / composting rose from 11% to 43% (now the largest) (1)	2
01.2	<i>Any 3 from the points below - 1 mark each, maximum 3.</i> • Organic matter decays anaerobically releasing methane (and CO₂) - greenhouse gases (1) • Toxic leachate may leak into groundwater / rivers from poorly managed sites (1) • Uses large areas of land; loss of habitats / farmland; contamination prevents later use (1) • Resource value of metals, glass, plastics is lost (1) • Transport noise / congestion / litter / pests / odour (1)	3
01.3	<i>Any 4 from the points below - 1 mark each, maximum 4.</i> • Polymer / clay liner to prevent escape of leachate (1) • Collection and treatment of leachate (1) • Collection of methane and use as a fuel (1) • Impermeable cap (clay and polymer) when complete to stop water entering (1) • Regular covering with soil (pests, odour); deodorising sprays (1) • Perimeter fence to trap wind-blown litter (1) • Separation and recording of waste types; reactive / flammable wastes not dumped together (1)	4
01.4	<i>1 mark.</i> • Mixed / contaminated / dirty wastes (plastics, dirty paper) are difficult and costly to separate and recycle but can be burnt with energy recovery (1)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 280-282; AQA spec 3.4.3.2.12.	

Question	Marking guidance	Marks
02.1	<i>Any 3 from the points below - 1 mark each, maximum 3.</i> • Volume of ash is much less than the original waste (less landfill) (1) • Heat used for district heating / electricity generation (energy recovery) (1) • No sorting / complicated management needed (1) • Destroys pathogens / flammable organic wastes converted to CO₂ and water (1) • Refuse-derived fuel can replace fossil fuels (eg cement kilns) (1)	3
02.2	<i>1 mark for each point.</i> • High temperature and rapid cooling reduce the formation of toxic dioxins (from organic wastes reacting with chlorine) (1) • Scrubber dissolves hydrogen chloride (from chlorine in PVC) which is then neutralised with alkali / limestone; also removes heavy metals for storage at high pH (1) • Wet waste needs extra fuel to burn and lowers combustion temperature; composting recovers its nutrient value (1) • Recycling conserves the resource value of metals, glass and plastics that would otherwise be lost (1)	4
02.3	<i>1 mark for each point.</i> • Ash may contain concentrated heavy metals / toxic residues (1) • Must be stored in solid form at high pH / hazardous waste landfill so the metals remain insoluble (1)	2
02.4	<i>Any 1 from the points below - 1 mark each, maximum 1.</i> • Fuel (natural gas) needed to burn wet / non-flammable wastes (1) • Cost of treating effluent gases (SO₂, NO_x, HCl, smoke) (1)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 283-284; AQA spec 3.4.3.2.12.	
03.1	<i>2 marks for each point (partial credit for a partly correct point).</i> • Asbestos: fibres become airborne when it disintegrates and are inhaled, causing asbestosis (scar tissue thickens alveoli) and mesothelioma; disposal: double-wrapped in heavy-duty polythene in a specialised, recorded landfill (secure, permanent, sealed storage) (2) • Cyanide: very toxic enzyme inhibitor (aerobic respiration) if inhaled / ingested; disposal: incineration so carbon and nitrogen are separated and oxidised to CO₂ and NO_x (much smaller problem) (2) • High-level radioactive waste: intense ionising radiation, heat generation, long half-lives; disposal: vitrification - mixed with molten glass in sealed stainless steel containers in ventilated concrete stores (2)	6
03.2	<i>1 mark.</i> • The fibres are bound / cannot become airborne so cannot be inhaled (1)	1
03.3	<i>1 mark for each point.</i> • Waste is mixed with a cement slurry and poured into containers of impermeable, unreactive material (1) • The solid cement immobilises the waste so it cannot leak / leach out (1)	2
03.4	<i>1 mark.</i> • Even if the glass shatters the waste stays encapsulated in the fragments; glass is more stable / resistant to heat and leaching over very long timescales (1)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 284-285, 303; AQA spec 3.4.3.2.12.	

Question	Marking guidance	Marks
04.1	<i>1 mark for each point.</i> • Instability / landslips on the hillside, especially after heavy rain - compaction, drainage, establishing vegetation whose roots bind the spoil, regrading slopes (1) • Acid drainage from oxidation of pyrites, with toxic metals - collect and treat leachate, add lime to raise pH (1) • Spontaneous combustion of residual coal / hydrocarbons - layers of fine-grained material to reduce air flow (1) • Lack of nutrients / poor plant growth - add topsoil, sewage sludge, fertiliser (1) • Unattractive topography - landscaping / reshaping (1) • Dust / heavy metals - cover and keep at high pH (1)	6
04.2	<i>1 mark for each point.</i> • So that plant roots cannot reach the toxic layer (1) • Otherwise toxins could be taken up and brought to the surface / dispersed via leaf litter and food chains (1)	2
04.3	<i>1 mark for each point.</i> • Steel slag is alkaline, producing unusual calcareous / high-pH soil conditions (1) • Supports uncommon plants such as cowslips / orchids that cannot compete on fertile neutral soils (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 282-283; AQA spec 3.4.3.2.12.	
05.1	<i>1 mark for each point.</i> • Affluent people buy more consumer goods / trivia / disposable items and throw items away rather than repairing them (1) • Built-in obsolescence: products designed to have short lives / wear out / go out of fashion so customers upgrade (1) • Disposable products (razors, lighters, pens) that must be replaced (1) • Over-packaging to make products attractive; also increases wastes from manufacture and mineral extraction (1)	4
05.2	<i>Any 3 from the points below - 1 mark each, maximum 3.</i> • Properties of the waste: toxicity, flammability, degradability, radioactivity (1) • Population density (collection costs) (1) • Land availability for landfill (1) • Availability of recycling technology / markets (1) • Legislation / regulatory framework (landfill tax) (1) • Public willingness to recycle / environmental awareness (1) • Processing costs / household income (1)	3
05.3	<i>1 mark for each point.</i> • Advantage: aerobic decomposition produces little methane; nutrients / organic matter returned to soil (circular economy); reduces landfill volume (1) • Disadvantage: needs separation / collection of clean organic waste; odour / pests / pathogens if poorly managed; low-value product; not suitable for mixed waste (1)	2
05.4	<i>1 mark.</i> • Many different materials are mixed together / contaminated so identification and separation are costly and the quality of recovered material is reduced (1)	1

Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 279-282; AQA spec 3.4.3.2.12.	
06.1	<i>Correct answer only.</i> • Aerobic decomposition produces little methane and returns nutrients to the soil (1)	1
06.2	<i>1 mark for each point.</i> • It may contain concentrated heavy metals / toxic residues (1) • It is stored in solid form at high pH / in hazardous-waste landfill so the metals stay insoluble (1)	2
06.3	<i>Any 2 from the points below - 1 mark each, maximum 2.</i> • Impermeable polymer / clay liner to stop leachate escaping (1) • Collection and treatment of leachate (1) • Collection of methane and use as fuel (1) • Daily covering of waste / final capping and restoration (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 279-285; AQA spec 3.4.3.2.12.	